

D895

Geometry for Secondary Mathematics Teaching

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A carpenter at a theater company is building a triangular stage piece with sides of 4 feet and 9 feet. The company performs six shows a year. Which length could be the third side?

- A. 10 feet
- B. 3 feet
- C. 5 feet
- D. 13 feet

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

During a garden-planning task at a school that just built raised beds, a student says a 20-foot fence will always enclose the same area no matter the rectangle's shape. The school has an active parent group. Which teacher move best addresses this?

- A. Have students compare a 1 x 9 and a 5 x 5 rectangle: both have perimeter 20, but areas 9 and 25.
- B. Agree, because perimeter and area are always linked.
- C. Tell the student the idea is wrong and give the formula for area.
- D. Have students measure the fence with a tape measure.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A student teacher at a large suburban school asks the class to prove that A(0, 0), B(3, 4), C(8, 4), D(5, 0) is a rhombus. The school has a new tablet program. Which argument is a valid proof?

- A. It looks like a rhombus on the tablet screen.
- B. AB is parallel to CD and BC is parallel to DA.
- C. By the distance formula, AB, BC, CD, and DA each equal 5.
- D. The diagonals bisect each other.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A school club designs a logo shaped like a regular pentagon to print on T-shirts. The shirts will be sold at a fall fundraiser. What is the smallest rotation that maps the logo onto itself?

- A. 108 degrees
- B. 5 degrees
- C. 72 degrees
- D. 36 degrees

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A regular polygon maps onto itself with a smallest rotation of 40 degrees. How many lines of symmetry does it have?

- A. 4
- B. 18
- C. 9
- D. 40

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A roof truss drawn for a shop class has a triangle with angles $A = 50$ degrees and $B = 70$ degrees. The shop just bought a new table saw. The side at C is extended to form an exterior angle at C. What is its measure?

- A. 240 degrees
- B. 120 degrees
- C. 110 degrees
- D. 60 degrees

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

An exterior angle of a triangle measures 130 degrees, and one remote interior angle is 45 degrees. What is the other remote interior angle?

- A. 50 degrees
- B. 85 degrees
- C. 175 degrees
- D. 135 degrees

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A landscaper who has won local design awards lays out a triangular flower bed with sides 6 m, 8 m, and 11 m. The bed will hold 40 tulip bulbs. Is the triangle acute, right, or obtuse?

- A. It is not a triangle.
- B. Obtuse
- C. Acute
- D. Right

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A triangle has sides 7, 8, and 9. Classify it by its angles.

- A. Acute
- B. Obtuse, using 7 as the longest side
- C. Obtuse
- D. Right

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

In a geometry class that meets in a room with new smartboards, students test the claim: if two angles are supplementary, they form a linear pair. The class has 32 students. What is the best way to settle the claim?

- A. Vote, since 32 students is a large enough sample.
- B. Accept it, because every linear pair is supplementary.
- C. Draw ten linear pairs; since all are supplementary, the claim is proven.
- D. Give a counterexample: two separate angles of 100 and 80 degrees that do not share a side.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A student checks a conjecture on six triangles in GeoGebra and it holds each time. What can be concluded?

- A. It is false, because only six cases were checked.
- B. It is proven, because six cases is enough.
- C. It is proven, because the software is exact.
- D. The conjecture is supported but not proven; a general proof is needed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

On a worksheet made for a summer school class, a student circles the angle drawn with the longer rays as the larger angle, even though both angles measure 40 degrees. The worksheet was printed in color. Which teacher move best addresses the misconception?

- A. Move on, since angle size will be taught again later.
- B. Explain that the larger-looking angle is always larger.
- C. Tell the student to always pick the angle with shorter rays.
- D. Have students measure both angles with a protractor, then extend and shorten the rays and measure again.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A carpenter at a theater company is building a triangular stage piece with sides of 4 feet and 9 feet. ~~The company performs six shows a year.~~ Which length could be the third side?

Target word: possible third side **Decides:** $5 < x < 13$.

A	Correct	10 is between 5 and 13.
B	Wrong stem number	$3 + 4 = 7$ is less than 9.
C	Absolute language	$4 + 5 = 9$ makes a flat, degenerate triangle.
D	Absolute language	$4 + 9 = 13$ is not greater than 13.

2. Answer A Concept 2

During a garden-planning task ~~at a school that just built raised beds,~~ a student says a 20-foot fence will always enclose the same area no matter the rectangle's shape. ~~The school has an active parent group.~~ Which teacher move best addresses this?

Target word: best teacher move **Decides:** Show same perimeter, different area.

A	Correct	A concrete counterexample shows area varies with shape.
B	Absolute language	They are independent measures.
C	Adjacent category	Telling without evidence rarely changes the idea.
D	Dropped element	Measuring the perimeter again does not address area.

3. Answer C Concept 3

A student teacher ~~at a large suburban school~~ asks the class to prove that A(0, 0), B(3, 4), C(8, 4), D(5, 0) is a rhombus. ~~The school has a new tablet program.~~ Which argument is a valid proof?

Target word: valid rhombus proof **Decides:** All four sides must be shown equal.

A	Camouflage	Appearance is not proof.
B	Dropped element	That proves only a parallelogram, not equal sides.
C	Correct	Four equal sides define a rhombus.
D	Confusion set	Also only proves a parallelogram.

4. Answer C Concept 4

A school club designs a logo shaped like a regular pentagon ~~to print on T-shirts. The shirts will be sold at a fall fundraiser.~~ What is the smallest rotation that maps the logo onto itself?

Target word: smallest rotation angle **Decides:** $360 / 5$.

A	Confusion set	That is the interior angle of a regular pentagon.
B	Wrong stem number	Uses the number of sides as the angle.
C	Correct	$360 / 5 = 72$.
D	Inverted operation	Halves the correct angle; 36 degrees does not map vertices onto vertices.

5. Answer C Concept 4 · transfer

A regular polygon maps onto itself with a smallest rotation of 40 degrees. How many lines of symmetry does it have?

Target word: solve for n **Decides:** $n = 360 / 40$.

A	Confusion set	Confuses 40 degrees with 4 lines.
B	Inverted operation	Doubles n; a regular n-gon has n lines, not 2n.
C	Correct	A regular 9-gon has 9 lines of symmetry.
D	Wrong stem number	Uses the angle as the count.

6. Answer B Concept 5

A roof truss drawn for a shop class has a triangle with angles A = 50 degrees and B = 70 degrees. ~~The shop just bought a new table saw.~~ The side at C is extended to form an exterior angle at C. What is its measure?

Target word: exterior angle **Decides:** Sum of the remote interior angles.

A	Inverted operation	Doubles the correct value.
B	Correct	$50 + 70 = 120$.
C	Dropped element	Uses $180 - 70$, dropping angle A.
D	Confusion set	That is the interior angle at C.

7. Answer B Concept 5 · transfer

An exterior angle of a triangle measures 130 degrees, and one remote interior angle is 45 degrees. What is the other remote interior angle?

Target word: solve for a remote angle **Decides:** $130 - 45$.

A	Confusion set	That is the adjacent interior angle, $180 - 130$.
B	Correct	$130 = 45 + x$, so $x = 85$.
C	Inverted operation	Adds instead of subtracting.
D	Wrong stem number	Uses $180 - 45$.

8. Answer B Concept 6

A landscaper ~~who has won local design awards~~ lays out a triangular flower bed with sides 6 m, 8 m, and 11 m. ~~The bed will hold 40 tulip bulbs.~~ Is the triangle acute, right, or obtuse?

Target word: classify by angle **Decides:** $36 + 64 = 100 < 121$.

A	Confusion set	$6 + 8 = 14 > 11$, so the triangle exists.
B	Correct	$a^2 + b^2 < c^2$.
C	Inverted operation	Reverses the inequality rule.
D	Dropped element	Right needs $a^2 + b^2$ exactly equal to c^2 .

9. Answer A Concept 6 · transfer

A triangle has sides 7, 8, and 9. Classify it by its angles.

Target word: classify, reversed case **Decides:** $49 + 64 = 113 > 81$.

A	Correct	$a^2 + b^2 > c^2$ with $c = 9$.
B	Dropped element	c must be the longest side.
C	Inverted operation	Reverses the inequality rule.
D	Wrong stem number	113 is not equal to 81.

10. Answer D Concept 7

In a geometry class ~~that meets in a room with new smartboards~~, students test the claim: if two angles are supplementary, they form a linear pair. ~~The class has 32 students.~~ What is the best way to settle the claim?

Target word: disprove with a counterexample **Decides:** One case breaks the claim.

A	Camouflage	Class size does not settle a mathematical claim.
B	Confusion set	That is the converse, which is true; the claim itself is false.
C	Inverted operation	Tests the converse with examples; examples cannot prove a claim.
D	Correct	They are supplementary but not a linear pair.

11. Answer D Concept 7 · transfer

A student checks a conjecture on six triangles in GeoGebra and it holds each time. What can be concluded?

Target word: what examples show **Decides:** Examples do not prove.

A	Inverted operation	A lack of proof is not a counterexample.
B	Absolute language	No finite number of examples proves a general claim.
C	Confusion set	Accurate examples are still examples, not a proof.
D	Correct	Confirming cases never cover every triangle.

12. Answer D Concept 8

On a worksheet ~~made for a summer school class~~, a student circles the angle drawn with the longer rays as the larger angle, even though both angles measure 40 degrees. ~~The worksheet was printed in color.~~ Which teacher move best addresses the misconception?

Target word: best move **Decides:** Angle measure is the opening, not ray length.

A	Dropped element	Leaves the misconception in place.
B	Absolute language	Appearance based on ray length is misleading.
C	Inverted operation	Replaces one wrong rule with its reverse.
D	Correct	Shows that ray length does not change the opening.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	1	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Triangle inequality	Each side of a triangle must be less than the sum of the other two and greater than their difference.	Allowing a side equal to the sum or difference of the other two.	Two sides known and possible values of the third.
2	Area and perimeter are independent	Shapes with the same perimeter can have different areas, and the reverse; for a fixed perimeter, a square rectangle has the greatest area.	Believing that a larger perimeter always means a larger area.	A student claim linking perimeter and area, or a fencing problem.
3	Coordinate proof of a rhombus	To prove a quadrilateral is a rhombus on the coordinate plane, show all four sides are equal with the distance formula (or show it is a parallelogram with perpendicular diagonals).	Stopping after showing opposite sides are parallel, which only proves a parallelogram.	Four vertices and a request to prove a specific special quadrilateral.
4	Rotational and reflection symmetry of regular polygons	A regular n-gon has n lines of symmetry and maps onto itself under rotations of multiples of $360/n$ degrees.	Using the interior angle as the rotation angle.	A logo or tile shaped like a regular polygon and a question about rotating it onto itself.
5	Exterior angle theorem	An exterior angle of a triangle equals the sum of the two remote (non-adjacent) interior angles.	Adding the adjacent interior angle, or treating the exterior angle as equal to one remote angle.	A triangle with one side extended.
6	Classifying triangles with the Pythagorean converse	With longest side c: if $a^2 + b^2 = c^2$ the triangle is right, if $a^2 + b^2 > c^2$ it is acute, and if $a^2 + b^2 < c^2$ it is obtuse.	Mixing up which inequality means acute and which means obtuse, or not using the longest side as c.	Three side lengths and a request to classify by angle.
7	Counterexamples and proof	One counterexample disproves a general claim; any number of examples that fit cannot prove it.	Treating several confirming cases, even in dynamic software, as a proof.	A conjecture tested on a few cases, or a request to disprove a statement.
8	Angle measure is the amount of turn	An angle's measure depends on the opening between the rays, not on how long the rays are drawn.	Thinking an angle drawn with longer rays is bigger.	Student work comparing angles drawn with rays of different lengths.

Course scope (public description): Geometry for Secondary Mathematics Teaching is a comprehensive course designed to equip future educators with the expertise to teach secondary-level geometry. This course delves into core geometric concepts, including geometric reasoning, transformations, congruence, similarity, and coordinate geometry, while emphasizing the development of critical thinking and problem-solving skills. Students will engage in dynamic instructional strategies and hands-on learning experiences, using tools like GeoGebra and Desmos to create interactive lessons that address common misconceptions and support diverse learners. By the end of the course, students will have the confidence and skills to design, implement, and refine geometry lessons aligned with educational standards and tailored to the needs of a diverse student population, preparing them to inspire and engage the next generation of mathematicians. This methods course will count towards 5 hours of the 50 total

clinical hours that learners will gain from their SCED methods courses.